

PS 4] Q4

$$u(x,t) = \sum_{n=0}^{\infty} b_n \sin(\lambda_n x) e^{-\lambda_n^2 t}$$

$$\lambda_n = \frac{\pi}{2} + n\pi, \quad n=0, 1, 2, \dots$$

* IC: $u(x,0) = 1$

$$\hookrightarrow u(x,0) = \sum_{n=0}^{\infty} b_n \sin(\lambda_n x) = 1$$

* Orthogonal trig fns:

$$\int_{-L}^L \sin(\lambda_n x) \sin(\lambda_m x) dx = \begin{cases} L \delta_{nm} & n \neq m \\ 0 & n = m \end{cases}$$

→ "gets rid of the sum"

ex: $\sum_{n=0}^{\infty} \alpha_n \delta_{n0} = \alpha_0$

$$\delta_{nm} = \begin{cases} 1, & n=m \\ 0, & n \neq m \end{cases}$$

$$= \alpha_0 \delta_{00} + \alpha_1 \delta_{10} + \alpha_2 \delta_{20} + \dots + \alpha_0 + 0$$

$$u(x,0) = \sum_{n=0}^{\infty} b_n \sin(\lambda_n x) = 1$$

$$\sin(\lambda_m x) \sum_{n=0}^{\infty} b_n \sin(\lambda_n x) = \sin(\lambda_m x)$$

$$\sum_{n=0}^{\infty} b_n \sin(\lambda_n x) \sin(\lambda_n x) = \sin(\lambda_m x)$$

$$n=0$$

* Integrate both sides:

• For Q4: $L = 1$ From B.C.s:

$$u(0, t) = 0$$

$$u_x|_{x=1} = 0$$



$$\int_0^1 \sum_{n=0}^{\infty} b_n \sin(\lambda_n x) \sin(\lambda_n x) dx = \int_0^1 \sin(\lambda_n x) dx$$

$$\sum_{n=0}^{\infty} b_n \int_0^1 \sin(\lambda_n x) \sin(\lambda_n x) dx = \int_0^1 \sin(\lambda_n x) dx$$

Aside: $\int_{-L}^L f(x) dx = \int_{-L}^0 f(x) dx + \int_0^L f(x) dx$

↑
"flip" $\int_{-L}^0 \rightarrow -\int_0^{-L}$

$$= -\int_0^{-L} f(x) dx + \int_0^L f(x) dx$$

↑
 $y = -x$ (change variables?)
 $dy = -dx$

• $y = 0 \leftarrow x = 0$
 $y = -(-L) = L$

$$= -\int_0^L f(y) (-dy) + \int_0^L f(x) dx$$

$$= \int_0^L f(y) dy + \int_0^L f(x) dx$$

↑
"dummy" $y \rightarrow x$

$$\int_0^L f(x) dx + \int_0^L f(x) dx = 2 \int_0^L f(x) dx$$

$$\boxed{\int_{-L}^L f(x) dx = 2 \int_0^L f(x) dx} = \begin{cases} \delta_{nm} \\ 0 \end{cases}$$

* $\sin(x)$: 2π



PDE: $0 < x < 1$
 $\sin(\lambda_n x)$

* For us: $f(x) = \sin(\lambda_n x) \sin(\lambda_m x)$

• Symmetric: If: $x \rightarrow -x$
 Integral stays the same

$x \rightarrow -x$: $\sin(\lambda_n(-x)) \sin(\lambda_m(-x))$
 $(-1)(-1) \sin(\lambda_n x) \sin(\lambda_m x)$
 $= \sin(\lambda_n x) \sin(\lambda_m x)$

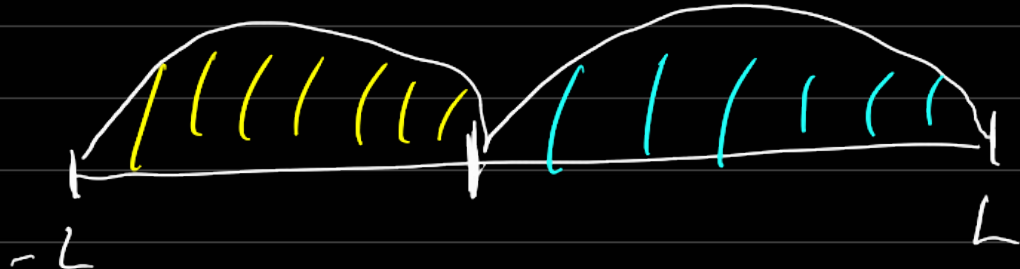
* $\int_0^1 \sin(\lambda_n x) \sin(\lambda_m x) dx = \frac{1}{2} \begin{cases} 1 \delta_{nm}, & n \neq m \\ 0, & \text{else} \end{cases}$

$\sum_{n=0}^{\infty} b_n \int_0^1 \sin(\lambda_n x) \sin(\lambda_m x) dx = \int_0^1 \sin(\lambda_m x) dx$

$= \sum_{n=0}^{\infty} b_n \left(\frac{1}{2} \delta_{nm} \right) = \int_0^1 \sin(\lambda_m x) dx$

$= \frac{1}{2} b_m = \int_0^1 \sin(\lambda_m x) dx$

aside:



from $-L$ to L : $||||| + |||||$

0 to L : $|||||$

$||||| = |||||$

$-L$ to L : twice that of 0 to L

$$\int_{-L}^L = 2 \int_0^L = \int_0^L = \frac{1}{2} \int_{-L}^L$$

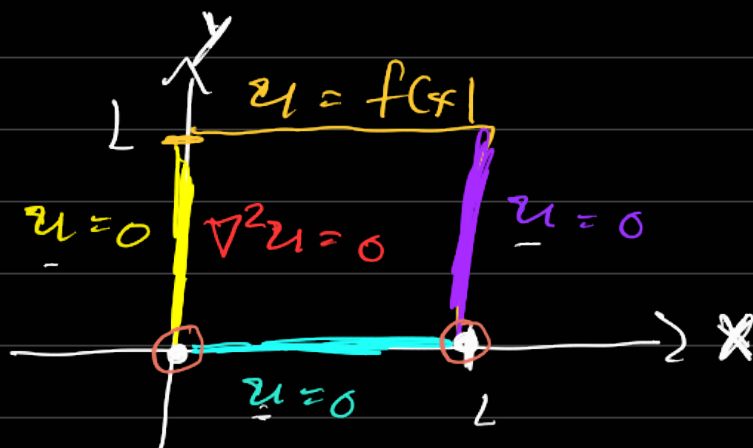
Q4: $b_m \frac{L}{2} = \int_0^L \sinh(\lambda_m x) dx$

Laplace Eq:

$$\nabla^2 u = 0$$

$$= u_{xx} + u_{yy}$$

$$\nabla^2 = \partial_x^2 + \partial_y^2$$



$$u(0, y) = 0$$

$$u(x, 0) = 0$$

$$u(L, y) = 0$$

$$u(x, L) = f(x)$$

$$0 < y < L$$

$$0 < x < L$$

$$0 < y < L$$

$$0 < x < L$$

$$\Delta = b^2 - 4Ac \leftarrow$$

$\Delta < 0$: Only BC

* Solve: using Sep of variables:

$$u(x, y) = X(x) Y(y)$$

plug into $z_{xx} + z_{yy}$

$$X Y'' + X'' Y = 0$$

$$\frac{X''}{X} + \frac{Y''}{Y} = 0$$

$$\boxed{\frac{X''}{X} = -\frac{Y''}{Y} = -\lambda^2}$$

two sets of ODEs:

(i) X :

$$X'' = -\lambda^2 X \quad *$$

(ii) Y :

$$Y'' = +\lambda^2 Y$$

For (i): $X(x) = \cancel{A} \cos(\lambda x) + B \sin(\lambda x)$

For (ii): $Y(y) = A \sinh(\lambda y) + B \cosh(\lambda y)$

$\hookrightarrow A = 0$: $x=0$: $\cos(0) = 1 \quad \checkmark$ BC
 $X(0) = \underline{A}$ $u(0, y) = 0$

$u(x, y) = X(x) Y(y)$ $\underline{A=0}$

For (i)

$$\boxed{X(x) = B \sin(\lambda x)}$$

For (ii):

$$Y(y) = A \sinh(\lambda y) + B \cosh(\lambda y)$$

Again

$$Y(0) : B = 0$$

$$u(x, 0) = 0$$

$$\boxed{u(x, y) = B \sin(\lambda x) \sinh(\lambda y)}$$

$$I(x) = A \sinh(\lambda y)$$

$$u(x, y) = AB \sin(\lambda x) \sinh(\lambda y)$$

• One more BC for λ :

Apply $u(L, y) = 0$

$$= AB \sin(\lambda L) \sinh(\lambda y) = 0$$

$$A, B, \sinh(\lambda y) \neq 0$$

$$\Rightarrow \sin(\lambda L) = 0 \quad \text{trivial}$$

$$\lambda_n = \frac{n\pi}{L}, \quad n = 0, 1, 2, \dots$$

$n = 1, 2, \dots$

$$u(x, y) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{L}x\right) \sinh\left(\frac{n\pi}{L}y\right)$$

"A.B"

One last BC to apply to solve for b_n :

$$u(x, L) = f(x)$$

$$u(x, L) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{L}x\right) \sinh(n\pi) = f(x)$$

(i) multiply by $\sin\left(\frac{m\pi}{L}x\right)$:

$$\sum_{n=1}^{\infty} b_n \sinh(n\pi) \left[\sin\left(\frac{n\pi}{L}x\right) \sin\left(\frac{m\pi}{L}x\right) \right] = \sin\left(\frac{m\pi}{L}x\right) f(x)$$

(ii) Integrate:

$$\sum_{n=1}^{\infty} b_n \sinh(n\pi) \left[\int_0^L \sin\left(\frac{n\pi}{L}x\right) \sin\left(\frac{m\pi}{L}x\right) dx \right] =$$

$$\int_0^L \sin\left(\frac{n\pi}{L}x\right) f(x) dx$$

$$\rightarrow = \frac{1}{2} (L \delta_{nm}) \quad n \neq m \neq 0$$

$$\sum_{n=1}^{\infty} b_n \sin(n\pi) \left[\frac{1}{2} L \delta_{nm} \right] = \int_0^L \sin\left(\frac{n\pi}{L}x\right) f(x) dx$$



$$b_m \sin(m\pi) \frac{L}{2} = \int_0^L \sin\left(\frac{m\pi}{L}x\right) f(x) dx$$

Steps this way until $f(x)$ is given

Solve for b_m :

$$b_m = \frac{2 \int_0^L \sin\left(\frac{m\pi}{L}x\right) f(x) dx}{L \sin(m\pi)}$$

$m \rightarrow n$

Laplace: $u(x, y) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{L}x\right) \sinh\left(\frac{n\pi}{L}y\right)$

$f(x)$: once given can "fully solve."

